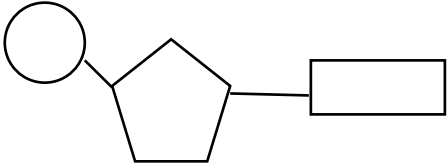
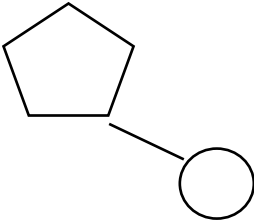
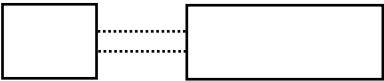


number		
1(a)(i)	B  <i>A is incorrect because the phosphate should be joined to C4</i> <i>C is incorrect because the base should be attached to C1</i> <i>D is incorrect because the phosphate should be joined to C4 and the base should be attached to C1</i>	(1)

number		
1(a)(ii)	B  <i>A is incorrect because a phosphate is attached to the sugar not the base</i> <i>C is incorrect because a phosphodiester joins a sugar with a phosphate, not two phosphates</i> <i>DC is incorrect because the phosphate is joined to the sugar not the base</i>	(1)

Question number	Answer	Mark
1(a)(iii)	C  <i>A is incorrect because complementary bases are joined by hydrogen bonds</i> <i>B is incorrect because one large base and one small base are joined together by hydrogen bonds</i> <i>D is incorrect because one large base and one small base are joined together</i>	(1)

Question number	Answer	Mark																								
1(b)(i)	<table><tr><th rowspan="2">Statement</th><th colspan="4">Correct statement about</th></tr><tr><th>both strands</th><th>only the complementary DNA strand</th><th>only the mRNA strand</th><th>neither strand</th></tr><tr><td>The number of guanines will be the same as in the template strand</td><td></td><td></td><td></td><td>X</td></tr><tr><td>The number of thymines will be the same as the number of adenines in the template strand</td><td></td><td>X</td><td></td><td></td></tr><tr><td>There will be no adenine present</td><td></td><td></td><td></td><td>X</td></tr></table>	Statement	Correct statement about				both strands	only the complementary DNA strand	only the mRNA strand	neither strand	The number of guanines will be the same as in the template strand				X	The number of thymines will be the same as the number of adenines in the template strand		X			There will be no adenine present				X	(3)
Statement	Correct statement about																									
	both strands	only the complementary DNA strand	only the mRNA strand	neither strand																						
The number of guanines will be the same as in the template strand				X																						
The number of thymines will be the same as the number of adenines in the template strand		X																								
There will be no adenine present				X																						

Question number	Answer	Additional guidance	Mark
1(b)(ii)	transcription (1)	IGNORE {post-transcription modification /protein synthesis} DO NOT ACCEPT {translation / reverse transcription}	(1)